A Three-Horizon AI Transformation for NAC Kazatomprom JSC

Capstone Executive Summary Report

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Executive Summary

NAC Kazatomprom JSC should execute a three-horizon AI transformation: (1) pilot continuous SCADA downhole telemetry and machine-learning acid dosing for 90 days on one wellfield block at the Inkai mine, to remove the manual sampling lag and reach a proposed 5%–6% reduction in sulfuric acid use per kg of uranium [6], [8]; (2) adapt AI multimodal freight tracking on the Trans-Caspian International Transport Route (TITR) to reach a proposed 10-day cut in average European transit time [9]; and (3) pilot one robotic drum-packing arm at the Ulba Metallurgical Plant to reach a proposed 30% cut in worker radiation exposure hours. Throughout, emergency well-block shutdowns, groundwater protection under the Subsoil Code, and verification of financial statements stay under human control [1], [7].

The evidence points to supply-side limits, not weak demand. Kazatomprom lowered its 2025 production expectations because sulfuric acid was hard to obtain [8]. A large share of Western shipments, about 48% in my Assignment 1 evidence, moved onto the TITR corridor [9]. In 2023, KZT 423,314 million of the company's assets sat in inventory [1]. Kazatomprom's owner, Samruk-Kazyna, holds 75% of the company and expects domestic investment and stable regional employment [1], which is why I recommend software and pilots, not foreign acquisitions.

The financial baseline is strong but not fully liquid. The audited 2023 figures show revenue of KZT 1,434,635 million, net profit of KZT 580,335 million (a 40.45% margin), a current ratio of 3.00 and a cash ratio of only 0.49 [1], [2]. Comparing with Cameco [3] shows which ideas transfer: automated process control, automated packaging and freight tracking can be evaluated for Kazatomprom, while Cameco's hard-rock mining methods and its 49% Westinghouse stake cannot.

The 12-month roadmap runs in three horizons: a 90-day telemetry pilot (Months 0–3), TITR tracking and the Ulba robot pilot (Months 4–8), and a scale-up decision (Months 9–12), each with a named owner and a decision gate. **Important:** the operating parameters (the 48–72 hour assay lag, the ±5% sign-off gate, the 12 bar escalation, the 6-hour manual fallback, the 20,000+ wellhead scale) and the three pilot KPIs are proposed operating-model design assumptions and pilot targets developed in Assignments 1 and 2. They are not verified historical facts published by Kazatomprom.

1. Business Context

In Week 1 I assumed that uranium demand would be the main business issue. After checking the sources, I revised this view. Three constraints bind Kazatomprom more tightly than demand. First, sulfuric acid is the key reagent in in-situ recovery (ISR) [6], and a domestic shortage led the company to lower its 2025 production expectations [8]. Second, sanctions risk on Russian routes pushed a large part of Western deliveries, about 48% in my Assignment 1 evidence, onto the TITR corridor through the Caspian Sea and Azerbaijan, which adds transfers, weather exposure and customs steps [9]. Third, working capital is tied up: at the end of 2023 inventory was KZT 423,314 million, of which KZT 328,015 million was uranium products [1]. The company is profitable, but its cash is slow to free up. For this reason my recommendation focuses on chemical efficiency, delivery time and liquidity, and not on selling more.

2. Environmental and Stakeholder Analysis

Samruk-Kazyna, the state wealth fund, owns 75% of Kazatomprom, and 25% is traded on the London and Astana exchanges [1]. The state owner wants dividends for the national budget, capital kept in Kazakhstan, and stable jobs in the southern regions where the mines are located. Public investors want returns and clear reporting. The Ministry of Energy enforces the Code of the Republic of Kazakhstan on Subsoil and Subsoil Use, which protects groundwater [7]. These three groups shape my design. The state's domestic focus leads to the rejection of foreign acquisitions in §3. The Subsoil Code means that shutdowns and groundwater escalations must stay human-only in §4. Investor reporting means that financial figures need a named reviewer, as set out in §6.

3. Business Model and Comparative Transfer Analysis

Kazatomprom uses ISR only. An acid solution is pumped through porous sandy ground, and the uranium-bearing liquid is brought to the surface [6]. This avoids moving ore and keeps costs low, but it makes the business sensitive to reagent supply and to sampling quality. Cameco mines hard rock underground at McArthur River and Cigar Lake and holds a 49% stake in Westinghouse Electric Company, which gives it a downstream position in the reactor business [3]. Table 1 states my verdict on each idea. Automated chemical process control and automated drum packaging are standard industrial practice in the nuclear fuel cycle, and I evaluated them for transfer; I did not assume Cameco uses the same equipment.

Table 1. Transfer verdicts from Cameco (A) to Kazatomprom (B)

Idea from Company A	Verdict	Reason
Foreign reactor-business M&A (49% Westinghouse stake) [3]	**Reject**	Conflicts with the state owner's aim of keeping capital in Kazakhstan [1].
Underground hard-rock mining automation [3]	**Reject**	Kazatomprom has no underground hard-rock mines; it uses liquid ISR in sandy ground [6].
Automated chemical process control and continuous telemetry	**Adopt / Pilot**	Fits ISR directly and targets the acid problem [6], [8].
Automated drum packaging	**Pilot**	One arm at Ulba to test lower worker radiation exposure.
Shipment tracking after the Inkai delay [3]	**Adapt**	Must be redesigned for the multi-border TITR corridor [9].

4. Consolidated Human–AI Operating Model

The model divides work into four tiers across four functions (Table 2). The aim is that AI handles volume and speed, and named people keep authority over safety, legal and financial decisions. The review points and fallbacks are design assumptions from Assignments 1 and 2, based on the principles of the course text [10].

Table 2. Four-tier task allocation, review points and fallbacks (proposed design)

Tier	Wellfield operations	Ulba processing	TITR logistics	Finance
Human only	Emergency well-block shutdown; groundwater decisions [7]	Nuclear safety and licence decisions	Contract renegotiation	Signing financial statements
Human + AI	Reserve-decline modelling and drill patterns	Packing schedule planning	Inventory swap and loan scheduling	Budget scenarios
AI executes + human review	Dosing alerts and draft valve orders; Lead Engineer signs any shift above ±5%	Fault alerts reviewed by shift supervisor	ETA variance above 24 h checked by Logistics Officer	Ratio calculations checked by Senior Analyst against IFRS statements
Automated	Telemetry collection and noise filtering across 20,000+ wellheads	Robotic drum filling inside the cell	Status data collection	Data entry matching

Tier	Wellfield operations	Ulba processing	TITR logistics	Finance
Fallback	Drift above 3%: 6-hour manual hydrometer rounds	Shielded manual tele-manipulator	Manual customs filing	Manual spreadsheet check

5a. Operations and Supply Chain

In the current state, I assume that samples are collected by hand across more than 20,000 wellheads and that laboratory results arrive after 48–72 hours. During that time valves stay in their old positions and acid may be overused. This lag is a design assumption from Assignment 2, not a published Kazatomprom figure. The proposed pilot installs continuous SCADA downhole telemetry on one wellfield block at Inkai for 90 days. Software filters sensor noise, flags dosing anomalies and drafts valve changes; the Lead Engineer approves any shift above ±5%. A pressure spike above 12 bar alerts the Environmental Safety Director so that the block can be halted. If a sensor drifts by more than 3%, the block goes back to manual 6-hour rounds. The pilot target is a 5%–6% reduction in acid use per kg of uranium. This is a proposed target, benchmarked against the IAEA manual [6] and Cameco's process-control practice [3], and not a result Kazatomprom has reported.

At Ulba I propose one robotic drum-packing arm. The target is a 30% reduction in worker radiation exposure hours over six months. This is also a proposed pilot target, to be measured against current dosimeter records. A robot fault reverts to the shielded manual tele-manipulator.

5b. Marketing and Customer

Nuclear utilities buy under long-term contracts, so reliability of delivery matters more than price changes. Cameco's 2023 Annual Report shows how a delay affects this: for the Inkai joint venture, only two-thirds of a shipment arrived in the fourth quarter, and the rest arrived later at a Canadian port. Cameco managed the gap through inventory buffers, long-term purchase agreements and loan arrangements [3]. Kazatomprom has a similar tool, an inventory loan arrangement with ANU Energy [1], but financing a delay is costlier than avoiding it. I propose adapting AI tracking to the TITR, which combines rail, Caspian Sea weather and several customs borders. The system predicts arrival times, and any variance above 24 hours is checked by the Logistics Officer against live carrier updates. A port closure longer than 72 hours goes to the Commercial Director, who can release inventory buffers. If the data feed fails, customs filing returns to manual brokerage. The target is a 10-day reduction in average European transit time; it is a proposed target, not an observed result [9].

5c. Workforce and Career Capability

The plan changes what people do, so training is part of the business case. The World Economic Forum's *Future of Jobs Report 2025* [4] and Green's 2024 OECD paper [5] both discuss how AI changes the demand for skills, including less demand for routine clerical tasks. I translate this into three capabilities for southern regional wellfield engineers and junior analysts. *Human* capability means ethical judgement, for example when deciding how to respond to a delay or a groundwater risk. *Meta* capability means adapting to change, such as a closed route. *Digital* capability means checking AI output: shipment arrival times, dosing alerts and financial ratios. The retraining would give engineers practice in questioning sensor alerts and give analysts practice in recalculating ratios from the audited statements. These findings are general labour-market evidence and not Kazatomprom data.

6. Governance, Verification and Assurance

Assurance applies to the whole model. **Privacy and security:** telemetry, shipment and financial data stay in company systems, and only named roles can approve changes. **Bias:** route recommendations are checked so that the system does not favour a corridor only because it has more historical data. **Reliability:** sensor drift above 3% ends automated dosing, and the block returns to manual rounds. **Accountability:** every AI-drafted action has a named human approver, and an action log records who approved what. **Safety:** a pressure spike above 12 bar escalates to the Environmental Safety Director, who can halt a block; this protects groundwater under the Subsoil Code [7]. **Fallbacks:** manual 6-hour hydrometer rounds for wellfields, a shielded manual

override at Ulba and manual customs filing for logistics. Financial figures are rechecked by the Senior Analyst against the audited statements, as shown in §7.

7. Business Case, KPIs and Assumption Verification

Table 3 shows the audited 2023 baseline. I calculated the ratios from the reported figures [1], [2].

Table 3. Audited 2023 baseline (KZT million unless stated)

Measure	Value	Calculation
Revenue	1,434,635	Reported [1], [2]
Net profit	580,335	Margin = 580,335 ÷ 1,434,635 = 40.45%
Current assets / current liabilities	1,290,336 / 430,812	Current ratio = 3.00
Inventory (of which uranium products)	423,314 (328,015)	Reported [1]
Cash and cash equivalents	211,912	Cash ratio = 211,912 ÷ 430,812 = 0.49
Operating cash flow	432,225	Reported [1], [2]

Week 8 discrepancy audit. My earlier working calculation used unverified figures of KZT 1,250.0 billion in current assets and KZT 310.5 billion in current liabilities, which gave a misleading current ratio of 4.02. After I checked the audited 2023 statements, the ratio became 3.00. The same check showed a cash ratio of only 0.49, because KZT 423,314 million is held in inventory awaiting physical delivery. The ANU Energy inventory loan arrangement points in the same direction [1]. The lesson is that the company is solvent, but cash is not as free as 4.02 suggested. For this reason I propose that the Senior Analyst rechecks ratios against the statements before any board paper.

I do not claim any monetary saving, because no verified cost data exist for the pilots. Table 4 separates what is verified from what is assumed.

Table 4. Assumption and verification table

Claim	Status	Basis and how it was checked
2023 revenue, profit, assets, liabilities, inventory, cash, cash flow	Verified	Audited figures in [1], [2]; ratios recalculated.
Samruk-Kazyna 75%, public float 25%	Verified	Shareholder information in [1].
Cameco hard-rock mines, 49% Westinghouse stake, Inkai delay	Verified	Cameco 2023 Annual Report [3].
Acid shortage, TITR rerouting	Reported	News reports [8], [9]; the 48% share comes from my Assignment 1 evidence.
48–72 h lab lag, 20,000+ wellheads, ±5%, 12 bar, 3%, 6 h	Design assumption	Proposed in Assignments 1 and 2; not published Kazatomprom data.
5%–6% less acid; 10 days faster; 30% less exposure	Pilot target	Proposed targets, benchmarked on [6] and [3]; to be tested in the pilots.
Pilot costs and savings	Not estimated	No verified data; quotes are required before approval.

8. 12-Month Implementation Roadmap

Table 5 sets out the plan in three horizons. Every KPI target is a proposed pilot target, and cost bases are described by resource type because no verified costs exist.

Table 5. Roadmap with owners, KPIs, risks and decision gates

1. Action	2. Owner	3. KPI (baseline → target)	4. Dependency	5. Cost / resource basis	6. Principal risk + control	7. Decision gate
Horizon 1: Months 0–3						

1. Action	2. Owner	3. KPI (baseline → target)	4. Dependency	5. Cost / resource basis	6. Principal risk + control	7. Decision gate
Install SCADA telemetry on one Inkai block	Inkai Operations Manager	Assay lag 48–72 h (assumed) → near real time	Sensor supply; IT support	Existing SCADA, pilot sensors; quote needed	Sensor drift; manual 6-hour rounds	Month 3: continue if data quality holds
Run ML dosing screening with ±5% gate	Lead Engineer	Acid per kg U: baseline → 5%–6% lower	Telemetry data	Engineer time	Wrong dosing; Lead Engineer sign-off	Month 3: scale or stop
Horizon 2: Months 4–8						
Adapt AI tracking on the TITR	Logistics Officer	Transit time: baseline → 10 days shorter	Carrier and customs data access	Software licence; quote needed	API outage; manual customs filing	Month 8: extend to all TITR shipments
Pilot one robotic packing arm at Ulba	Plant Safety Manager	Exposure hours: baseline → 30% lower	Safety approval	One robot; quote needed	Robot fault; manual tele-manipulator	Month 8: approve more arms
Horizon 3: Months 9–12						
Scale telemetry to more blocks	Chief Technical Officer	Blocks covered: 1 → agreed number	Pilot results	Sensors and training time	Over-trust in AI; mandatory review gate	Month 12: board decision
Train staff and recheck ratios quarterly	HR Lead; Senior Analyst	Staff trained; zero unreconciled figures	Pilot results	Training hours	Skill gaps; assessment	Month 12: annual review

9. Strategic Transfer Stress-Test Readiness

I tested the plan against three shocks. **Sensor drift or failure.** What still holds: the human-only shutdown rule and the 12 bar escalation. What changes: the affected block returns to 6-hour manual hydrometer rounds, so the acid target is paused for that block. What is abandoned: automated dosing on that block until the Lead Engineer confirms the data are reliable. **Caspian corridor or customs shock.** What still holds: the 24-hour variance check and the escalation to the Commercial Director after 72 hours. What changes: shipments use inventory buffers and the ANU Energy arrangement [1], and customs filing goes manual. What is abandoned: the 10-day target for the affected period, since delays outside the company's control cannot be tracked away. **Engineer capability shortage.** What still holds: the human-only tier and the sign-off gate, which require certified engineers. What changes: the pilot is limited to one block and training is brought forward. What is abandoned: scale-up in Months 9–12 until enough staff can review AI alerts. In each case I would rather reduce the scope than let AI act without a qualified reviewer.

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